

FITBUDDY AI

3. Project Design Phase

PROJECT REPORT DOCUMENT

Project: FitBuddy AI — Fitness Plan Generator

Group 1

M. Vijayasathy — Team Leader

Bharath

Harsha

Sumith

Arun Vasan

Objective

To translate the requirements into a modular architecture, data model, user flow and interface design.

Prepared for academic project submission

3. Project Design Phase

System Architecture

- Browser UI → FastAPI HTML routes → Jinja2 templates.
- FastAPI routes → AI service → Google Gemini API (or mock response in local test mode).
- FastAPI routes → database access layer → SQLite database via SQLAlchemy.
- JSON clients → REST API under /api/v1.

Major Modules

- main.py: initializes FastAPI, database lifecycle and routers.
- routes.py: web form handlers, feedback handling, admin protection and API endpoints.
- ai_service.py: prompts, Gemini calls, mock responses and plan formatting.
- schemas.py: Pydantic validation for user and feedback data.
- database.py / crud.py: persistence and database operations.
- templates/ and static/: user interface and styling.

Data Model

- User entity: unique user_id, name, age, weight, goal, intensity, experience_level, created_at and updated_at.
- Plan entity: user_id foreign key, original_plan, updated_plan, nutrition_tip, feedback, is_current, created_at and updated_at.
- Relationship: one user may have multiple plan records; only the latest active plan is marked current.

User Flow

- 1. Open home page. 2. Enter profile and preferences. 3. Submit form. 4. Validate input and save profile. 5. Generate workout and wellness tip. 6. Save plan and display result. 7. User submits feedback. 8. AI revises plan, database stores revision, and updated result is displayed.

API Design

- GET /api/v1/health — service and AI configuration status.
- POST /api/v1/plans — create a plan from a JSON profile.
- GET /api/v1/plans/{user_id} — retrieve profile and plan record.
- POST /api/v1/plans/{user_id}/feedback — revise plan using feedback.

Interface Design

- Home page: short introduction and fitness profile form. Result page: user details, formatted seven-day plan, wellness tip and feedback form. Error page: clear message. Admin page: protected listing of users and current plans.

Design Decisions

- Use a modular monolithic architecture for straightforward development; SQLite for local persistence; server-rendered Jinja2 pages for a simple web interface; REST API for programmatic access; configurable AI model names and mock mode for testability.

Design Limitation

- Generated plans depend on model output and user-provided information. The application is not a substitute for professional fitness, nutrition or medical advice.

Project Team — Group 1

- M. Vijayasathy — Team Leader
- Bharath
- Harsha
- Sumith
- Arun Vasan